SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I S. Devaganesh (191126014) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Devaganesh

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

CLOUD Computing

DEFINITION

Internet-based Computing

Store data and use applications online

CHARACTERISTICS

on-demand Service

Resource pooling

Rapid scaling

Measured use

BENEFITS

Cost saving

Flexibity

Scalability

Easy access

LIMITATIONS

Security risks

Internet dependecy

Downtime

Privacy issues

Evolution of Cloud Computing

- Hardware Evolution
  - Mainframes
  - Personal pcs
  - virtual servers
- Internet Software Evolution
  - Web browsers
  - Web applications
  - Internet access
- Distributed Systems
  - Resource sharing
  - Multiple Computers
  - Distributed storage

Cloud Computing

Service Delivery

Iaas - Paas - Saas

on demand Internet Service

CLOUD SERVICE MODELS

IaaS

Infrastructure as a Service

PROVIDER:

- Servers
- Storages
- Networking

USER

- OS and apps
- Data
- Configurations

PaaS

Platform as a Services

PROVIDER:

- Infrastructure
- OSs and runtime
- Development platform

USER

- Application development
- Application data

SaaS

Software as a Services.

PROVIDER

- Complete app
- Maintenance
- updates

USER

- Uses software
- Manages own data

CLOUD COMPUTING ENBLING TECHNOLOGIES

VIRTUALIZATION

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Creates Virtual resource

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- virtual machines
- virtual storage
- Virtual networks

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BENEFITS

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- Resource sharing
- Cost reduction
- Fast provisioning

WEB SERVICES

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Connects applications over networks.

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- APIs
- HTTP/HTTPs
- data exchange

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BENEFITS

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- Easy integration
- Communication
- Interoperability

14/10/26